



**Barker College**

**2001  
TRIAL  
HIGHER SCHOOL  
CERTIFICATE**

# **Mathematics Extension 2**

**AM WEDNESDAY 8 AUGUST**

## **General Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Make sure your Barker Student Number is on ALL pages
- Board-approved calculators may be used
- A table of standard integrals is provided on page 10.
- ALL necessary working should be shown in every question

## **Total marks (120)**

- Attempt Questions 1 – 8
- All questions are of equal value

**Total marks (120)**

**Attempt Questions 1 – 8**

**ALL questions are of equal value**

Answer each question on a SEPARATE sheet of paper

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**Marks**

**Question 1 [15 marks] [START A NEW PAGE]**

(a) Find

$$\int \frac{dx}{x^2 - 16x + 80}$$

**2**

(b) Evaluate

(i)  $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin \theta} d\theta$

**3**

(ii)  $\int_0^2 \frac{8 dx}{(x + 2)(x^2 + 4)}$

**4**

(iii)  $\int_0^{\pi} e^x \cos x dx$

**3**

(c) Find  $\int \frac{2x}{\sqrt{4x - x^2}} dx$

You may wish to use the substitution of  $u = x - 2$ .

**3**

**Question 2 [15 marks] [START A NEW PAGE]**

(a) (i) Find all real numbers  $x$  and  $y$  such that  $(x + iy)^2 = -3 + 4i$ . 2

(ii) Hence, solve the equation  $z^2 - 3z + (3 - i) = 0$ . 1

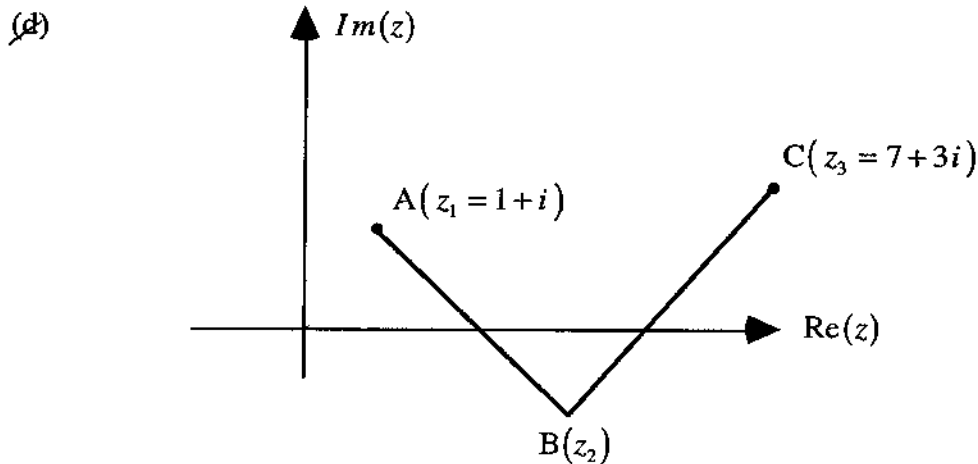
(b) (i) Express  $\sqrt{3} + i$  and  $\sqrt{3} - i$  in modulus-argument form. 2

(ii) Hence, simplify  $(\sqrt{3} + i)^{15} + (\sqrt{3} - i)^{15}$  1

(c) Sketch the locus specified by

(i)  $|z| \leq |z - 2|$  and  $-\frac{\pi}{4} \leq \arg z \leq \frac{\pi}{4}$  3

(ii)  $\operatorname{Re}\left(z - \frac{1}{z}\right) = 0$  (State the equation(s) of the locus). 4



The points A and C represent the complex numbers

$$z_1 = 1 + i \text{ and } z_3 = 7 + 3i$$

Find the complex number  $z_2$  represented by B such that  $\triangle ABC$  is isosceles and right angled at B.

**2**

**Question 3 [15 marks] [START A NEW PAGE]**

(a) If  $f(x) = (x - 1)(x - 3)$  then sketch

(i)  $y = \frac{1}{f(x)}$  2

(ii)  $y = f(|x|)$  2

(iii)  $|y| = f(x)$  2

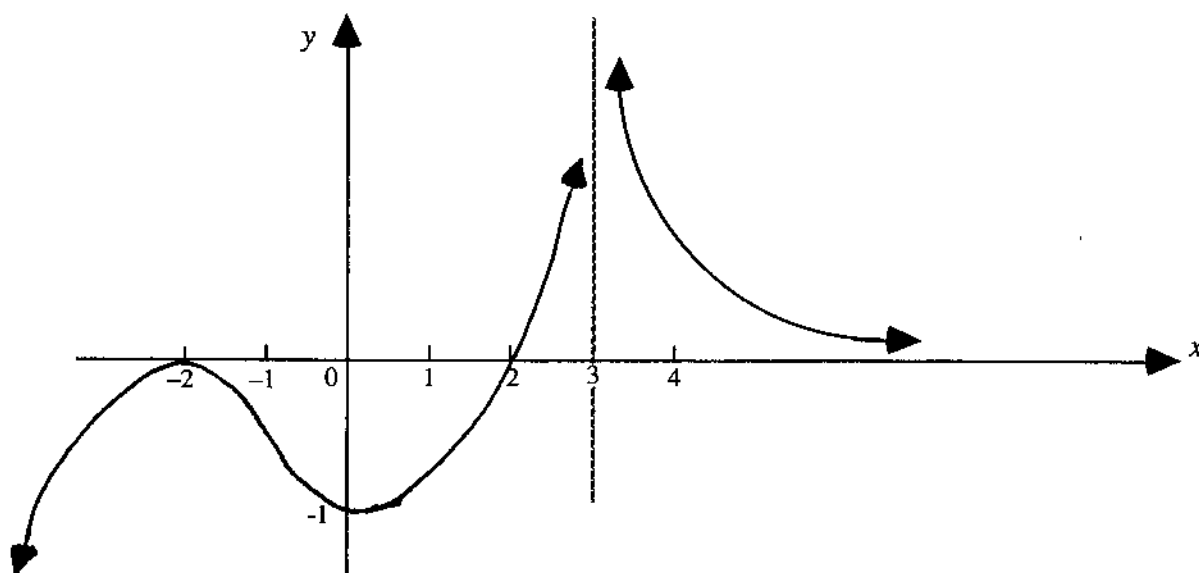
(b) (i) Find the stationary points and the asymptotes of the function

$$y = \frac{(x + 1)^4}{x^4 + 1}$$
 2

(ii) Sketch this function labelling all essential features. 1

(iii) Use the graph to find the set of values of  $k$  for which  $(x + 1)^4 = k(x^4 + 1)$  has two distinct real roots. 2

(c) Given the graph of  $y = f'(x)$  below, sketch the graph of  $y = f(x)$ .  
 $y = f'(x)$  is the derivative of  $y = f(x)$ . 4



**Question 4 [15 marks] [START A NEW PAGE]**

(a) An ellipse has the equation  $\frac{x^2}{16} + \frac{y^2}{9} = 1$

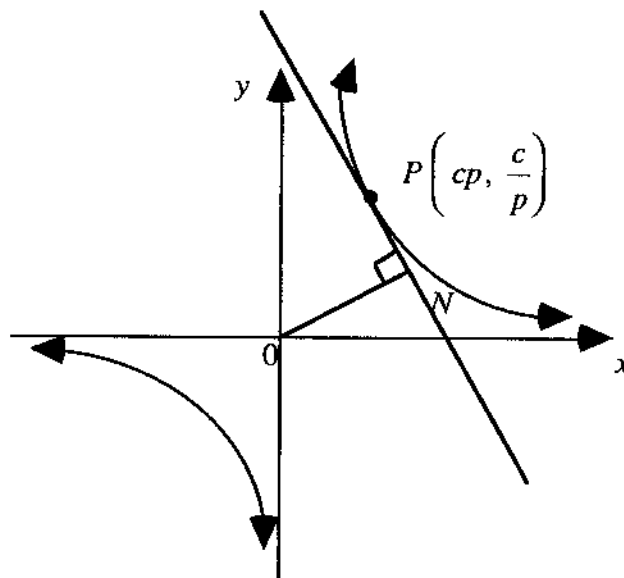
(i) Sketch the ellipse showing the foci  $S$  and  $S'$  and the directrices. 4

(ii) Prove that the tangent at the point  $P(4\cos\theta, 3\sin\theta)$  to the ellipse has the equation  $\frac{x\cos\theta}{4} + \frac{y\sin\theta}{3} = 1$  3

(iii) The ellipse meets the  $y$ -axis at  $B$  and  $B'$ . The tangents at  $B$  and  $B'$  meet the tangent at  $P$  at the points  $Q$  and  $Q'$ .  
Prove that  $BQ \cdot B'Q' = 16$  3

(b) The line through  $O$  perpendicular to the tangent at  $P\left(cp, \frac{c}{p}\right)$  on the rectangular hyperbola  $xy = c^2$  meets the tangent at  $N$ .

Find the coordinates of  $N$  and show that as  $p$  varies, the locus of  $N$  is  $(x^2 + y^2)^2 = 4c^2xy$ . 5



**Question 5 [15 marks] [START A NEW PAGE]**

(a) A solid has as its base the ellipse  $\frac{x^2}{36} + \frac{y^2}{16} = 1$

If each section perpendicular to the major axis is an equilateral triangle, show that the volume of the solid is  $128\sqrt{3}$  units<sup>3</sup>.

4

(b) The region  $(x - 2R)^2 + y^2 \leq R^2$  is rotated about the y-axis forming a solid of revolution called a torus.

By summing volumes of cylindrical shells, show that the volume of the torus is  $4\pi^2 R^3$  units<sup>3</sup>.

6

(c) The angles of elevation of the top of a tower P measured from three points A, B, C are  $\alpha$ ,  $\beta$ ,  $\gamma$  respectively.  
A, B, C are in a straight line such that  $AB = BC = a$ , but the line AC does not pass through S, the base of the tower.

(i) If  $\angle ABS = \theta$ , show that

$$(CS)^2 = a^2 + h^2 \cot^2 \beta + 2ah \cot \beta \cos \theta$$

2

(ii) Prove that the height of the tower is

$$\frac{a\sqrt{2}}{\{\cot^2 \alpha + \cot^2 \gamma - 2\cot^2 \beta\}^{\frac{1}{2}}}$$

3

**Question 6 [15 marks] [START A NEW PAGE]**

- (a) Given that  $a, b$  and  $c$  are the roots of the equation  $x^3 + qx + r = 0$ , find the cubic equation in  $y$ , in terms of  $q$  and  $r$ , whose roots are  $(b + c - 2a), (c + a - 2b)$  and  $(a + b - 2c)$

3

- (b) Using  $\tan 3\theta = \tan(2\theta + \theta)$ , show that

$$\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

2

- (ii) Find the value of  $x$  for which  $3 \tan^{-1} x = \frac{\pi}{2} - \tan^{-1}(3x)$  where  $\tan^{-1} x$  and  $\tan^{-1}(3x)$  both lie between  $0$  and  $\frac{\pi}{2}$

4

- (c) Using mathematical induction, prove that

$$\sum_{r=0}^n \frac{1}{(r+1)(r+2)(r+3)} = \frac{(n+1)(n+4)}{4(n+2)(n+3)}$$

4

- (iii) Hence, find  $\lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{1}{(r+1)(r+2)(r+3)}$

2

**Question 7 [15 marks] [START A NEW PAGE]**

(a) ~~(i)~~ Given that  $\int_0^\pi \sin^n \theta d\theta = I_n$ , prove that

$$nI_n = (n - 1)I_{n-2}$$

**3**

~~(ii)~~ Hence, evaluate  $I_8$

**1**

~~(iii)~~ Use the result  $\int_0^a f(x)dx = \int_0^a f(a - x)dx$  to show that

$$\int_0^\pi x \sin^n x dx = \left(\frac{\pi}{2}\right) I_n$$

**2**

(b) ~~(i)~~ Use De Moivre's Theorem to prove that, if  $2\cos\theta = x + \frac{1}{x}$ ,  
then  $2\cos n\theta = x^n + \frac{1}{x^n}$

**1**

~~(ii)~~ Hence, or otherwise, solve the equation

$$5x^4 - 11x^3 + 16x^2 - 11x + 5 = 0$$

**4**

(c) At the ends of three successive seconds, the distances of a point moving with Simple Harmonic Motion from its mean position, measured in the same direction, are 1, 5 and 5 metres.

Show that the period of the complete oscillation is  $\frac{2\pi}{\cos^{-1}\left(\frac{3}{5}\right)}$  seconds.

**4**

## Question 8 [15 marks] [START A NEW PAGE]

- (a) A ball thrown from a point  $P$  with velocity  $V$ , at an inclination  $\alpha$  to the horizontal reaches a point  $Q$  after  $t$  seconds.  
Show that if  $PQ$  is inclined at  $\theta$  to the horizontal, (where  $\alpha > \theta$ ), then the direction of motion of the ball, when at  $Q$ , is inclined to the horizontal at an acute angle of  $\tan^{-1}[2 \tan \theta - \tan \alpha]$ .

You may use the result without proof

$$x = V \cos \alpha \times t$$

$$y = V \sin \alpha \times t - \frac{1}{2} g t^2$$

4

- (b) (i) A gun fires shells with muzzle velocity  $V$ . Ignoring air resistance, show that the range on a horizontal plane is  $\frac{V^2 \sin 2\theta}{g}$  where  $\theta$  is the angle of elevation of the gun and  $g$  is the acceleration due to gravity.

2

- (ii) The gun and the target lie on the same horizontal plane. The gun fires, in the correct vertical plane, at the target using an angle of elevation  $\alpha$  and the shell falls short by a distance  $p$ . When the angle of elevation is changed to  $\beta$ , the shell overshoots the target by a distance  $q$ .

$$\text{Show that } \sin 2\theta = \frac{p \sin 2\beta + q \sin 2\alpha}{p + q}$$

4

- (c) Fred has three uniform tetrahedra (triangular pyramids). Each of these tetrahedra has one face black, one face white, one face red and one face green.  
When tossed onto a table, three faces of each tetrahedron can be seen.  
If the probability of any coloured face not being seen is equally likely, what is the probability that

- (i) no black face can be seen?

1

- (ii) exactly 2 black faces can be seen?

1

- (iii) at least 2 red faces can be seen?

1

- (iv) 3 white faces and only 1 green face can be seen?

2

END OF PAPER

Question 1

(a)  $\int \frac{dx}{x^2 - 16x + 80}$   
 $= \int \frac{dx}{(x-8)^2 + 16}$  ✓  
 $= \frac{1}{4} \tan^{-1} \left( \frac{x-8}{4} \right) + C$  ✓

(b) ii)  $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin \theta} d\theta$   
 $= \int_0^1 \frac{2}{(t+1)^2} dt$  ✓ let  $\tan \frac{\theta}{2} = t$   
 $= -2 \left[ \frac{1}{t+1} \right]_0^1$  ✓  $\frac{dt}{d\theta} = \frac{1}{2}(1+t^2)$   
 $= 1$  ✓

(ii)  $\int_0^2 \frac{8 dx}{(x+2)(x^2+4)}$   
 $\frac{1}{(x+2)(x^2+4)} = \frac{A}{x+2} + \frac{Bx+C}{x^2+4}$   
 $A = \frac{1}{8}, B = -\frac{1}{8}, C = \frac{1}{4}$  ✓  
 $I = \int_0^2 \left( \frac{1}{8(x+2)} + \frac{-\frac{1}{8}x + \frac{1}{4}}{x^2+4} \right) dx$   
 $= \left[ \ln(x+2) \right]_0^2 - \frac{1}{16} \int_0^2 \frac{x}{x^2+4} dx$  ✓  
 $+ 2 \int_0^2 \frac{1}{x^2+4} dx$   
 $= \ln 2 - \frac{1}{32} \left[ \ln(x^2+4) \right]_0^2 + \left[ \tan^{-1} \left( \frac{x}{2} \right) \right]_0^2$  ✓  
 $= \frac{\pi}{4} + \frac{1}{2} \ln 2$  ✓

(iii)  $I = \int_0^{\pi} e^x \cos x dx$   
 $= e^x \cos x + \int e^x \sin x dx$  ✓  
 $= e^x \cos x + [\sin x e^x - \int e^x \cos x dx]$   
 $= e^x \cos x + \sin x e^x - I$  ✓

$\therefore 2I = \left[ e^x (\cos x + \sin x) \right]_0^{\pi}$

$I = \frac{1}{2} [-e^{\pi} - 1]$  ✓

(c)  $\int \frac{2x}{\sqrt{4x-x^2}} dx$  let  $u = x-2$   
 $du = dx$   
 $= \int \frac{2(u+2)}{\sqrt{4(u+2)-(u+2)^2}} du$  ✓  
 $= 2 \int \frac{u+2}{\sqrt{4-u^2}} du$   
 $= 2 \int \frac{u}{\sqrt{4-u^2}} du + 4 \int \frac{1}{\sqrt{4-u^2}} du$   
 $= -2\sqrt{4-u^2} + 4 \sin^{-1} \left( \frac{u}{2} \right)$  ✓  
 $= -2\sqrt{4x-x^2} + 4 \sin^{-1} \left( \frac{x-2}{2} \right) + C$  ✓

Question 2

(a)  $(x+iy)^2 = -3+4i$   
 $x^2 - y^2 + 2xyi = -3+4i$   
 $xy = 2$  ✓  
 $x^2 - y^2 = -3$  ✓  
 solving simultaneous.  
 $x=2, y=1$  ✓  
 $x=-1, y=-2$  ✓

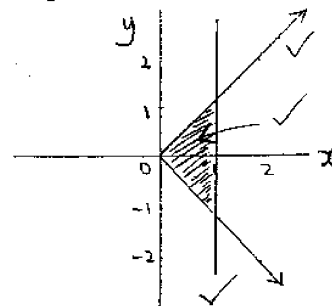
(b)  $z^2 - 3z + (3-i) = 0$   
 $z = \frac{3 \pm \sqrt{9-4(3-i)}}{2}$   
 $= \frac{3 \pm \sqrt{-3+4i}}{2}$   
 $= 2+i \text{ or } 1-i$  ✓

Question 2

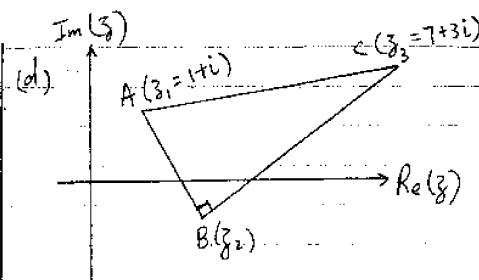
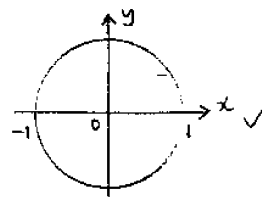
(b) (i)  $\sqrt{3}+i = 2 \left( \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$  ✓  
 $\sqrt{3}-i = 2 \left( \cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right)$  ✓

(ii)  $(\sqrt{3}+i)^{15} + (\sqrt{3}-i)^{15}$   
 $= 2^{15} \left( \cos \frac{15\pi}{6} + i \sin \frac{15\pi}{6} \right)$   
 $+ 2^{15} \left( \cos \frac{15\pi}{6} - i \sin \frac{15\pi}{6} \right)$   
 $= 2 \times 2^{15} \cos \frac{15\pi}{6}$   
 $= 0$  ✓

(c) (i)



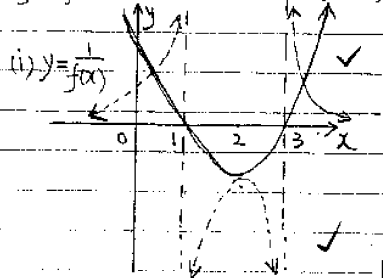
(ii)  $\operatorname{Re} \left( z - \frac{1}{z} \right) = 0$   
 $\operatorname{Re} \left( x+iy - \frac{x-iy}{x^2+y^2} \right) = 0$  ✓  
 $x - \frac{x}{x^2+y^2} = 0$   
 $x^2 + y^2 = 1$  ✓  
 or  $x=0$  ✓



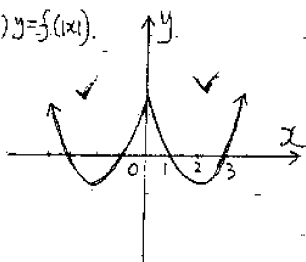
$z_1 = x+iy$   
 $i \vec{BC} = \vec{BA}$   
 $i[(7-x) + (3-y)i] = (1-x) + i(1-y)$   
 $-(3-y) + i(7-x) = (1-x) + i(1-y)$   
 Equating real part,  
 $-3+y = 1-x$   
 $x+y = 4$  — (1)  
 Equating imaginary part,  
 $7-x = 1-y$   
 $-x+y = -6$  — (2)  
 solving (1) + (2) simult.  
 $x=5, y=-1$   
 $z_2 = 5-i$  ✓

### Question 3

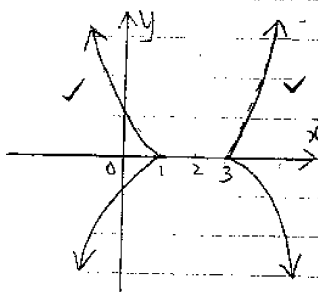
(a)  $f(x) = (x-1)(x-3)$



(ii)  $y = f(x)$



(iii)  $|y| = f(x)$



(b) (i)  $y = \frac{(x+1)^4}{x^4+1}$

$$\frac{dy}{dx} = \frac{(x^4+1)4(x+1)^3 - (x+1)^4 4x^3}{(x^4+1)^2}$$

$$= \frac{4(x+1)^3(1-x^3)}{(x^4+1)^2}$$

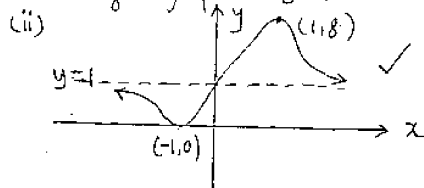
$y' = 0 \Rightarrow x = -1, 1$

Stationary points are

$(-1, 0), (1, 8)$

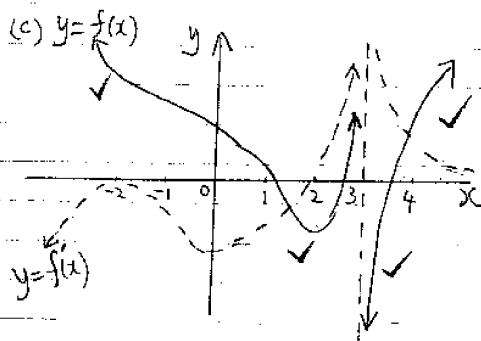
As  $x \rightarrow \infty, y \rightarrow 1$

Horizontal asymptote is  $y = 1$



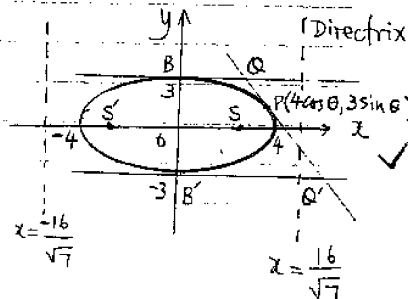
(iii)  $(x+1)^4 = K(x^4+1)$  has two distinct real roots

$\therefore$  the line  $y = K$  must meet the curve at two points  
 $0 < K < 1$  and  $1 < K < 8$



### Question 4

(a) (i)  $\frac{x^2}{16} + \frac{y^2}{9} = 1$



$b^2 = a^2(1-e^2)$

$9 = 16(1-e^2)$

$e = \frac{\sqrt{7}}{4}$

$S(ap, 0) = (\sqrt{7}, 0)$

$S'(-\sqrt{7}, 0)$

Directrix  $x = \pm \frac{a}{e}$   
 $= \pm \frac{16}{\sqrt{7}}$

(ii)  $\frac{dy}{dx} = \frac{-9x}{16y}$   
 $= -\frac{3 \cot \theta}{4}$

Equation of the tangent at P

$y - 3 \sin \theta = \frac{-3 \cot \theta}{4} (x - 4 \cos \theta)$

$\frac{y \sin \theta}{3} + \frac{x \cos \theta}{4} = 1$

(iii) At Q,  $y = 3$

$\therefore \frac{3 \sin \theta}{3} + \frac{x \cos \theta}{4} = 1$

$\therefore x_Q = \frac{4(1 - \sin \theta)}{\cos \theta}$

At Q',  $y = -3$

$\therefore \frac{-3 \sin \theta}{3} + \frac{x \cos \theta}{4} = 1$

$x_{Q'} = \frac{4(1 + \sin \theta)}{\cos \theta}$

BQ x B'Q'

$= \frac{4(1 - \sin \theta)}{\cos \theta} \times \frac{4(1 + \sin \theta)}{\cos \theta} = 16$

(b)  $y = \frac{c^2}{x}$   
 $y' = \frac{-c^2}{x^2}$

Eg of tangent at P,

$y - \frac{c}{p} = \frac{-c^2}{p^2} (x - cp)$

$x + p^2 y = 2cp$  (1)

Equation of ON is

$y = p^2 x$  (2)

Solving (1), (2) Simult to find coord N

$x_N = \frac{2cp}{1+p^4}$

$y_N = \frac{2cp^3}{1+p^4}$

using  $x(1+p^4) = 2cp$  and  $\frac{y}{x} = p^2$ ,

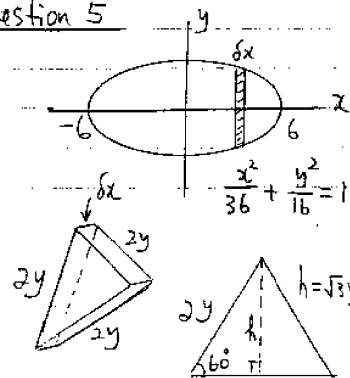
$x^2(1+p^4)^2 = (2cp)^2$

$x^2(1+\frac{y^2}{x^2})^2 = 4c^2(\frac{y}{x})$

$(x^2+y^2)^2 = 4c^2xy$

### Question 5

(a)



cross-sectional area of  $\Delta = \frac{1}{2} \sqrt{3} y^2$

$$\delta V = A \delta x$$

$$= \sqrt{3} y^2 \delta x \quad \checkmark$$

Vol of the solid

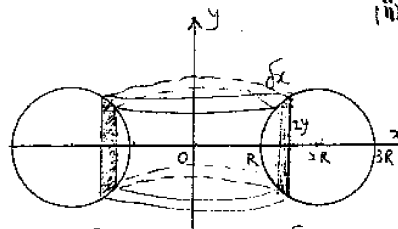
$$= \int \sqrt{3} y^2 dx$$

$$= \sqrt{3} \int_{-6}^6 16 \left(1 - \frac{x^2}{36}\right) dx \quad \checkmark$$

$$= 16\sqrt{3} \left[ x - \frac{x^3}{108} \right]_{-6}^6 \quad \checkmark$$

$$= 128\sqrt{3} \text{ units}^3 \quad \checkmark$$

(b)



$$\delta V = 2\pi x \cdot 2y \delta x$$

$$= 4\pi xy \delta x \quad \checkmark$$

$$\text{Volume} = \int_R^{3R} 4\pi x \sqrt{R^2 - (x-2R)^2} dx \quad \checkmark$$

$$\text{let } x-2R = R \sin \theta$$

$$dx = R \cos \theta d\theta$$

volume of the torus

$$= 4\pi \int_{-\pi/2}^{\pi/2} \sqrt{R^2 - R^2 \sin^2 \theta} \cdot (2R + R \sin \theta) R \cos \theta d\theta \quad \checkmark$$

$$= 4\pi R^3 \int_{-\pi/2}^{\pi/2} (2 \cos^2 \theta + \cos^2 \theta \sin \theta) d\theta$$

$$= 4\pi R^3 \left[ (\theta + \sin 2\theta) - \frac{1}{3} \cos^3 \theta \right]_{-\pi/2}^{\pi/2} \quad \checkmark \checkmark$$

$$= 4\pi R^3 \quad \checkmark$$

(c)

$$\begin{cases} AS = h \cot \alpha \\ BS = h \cot \beta \\ CS = h \cot \gamma \end{cases} \quad \checkmark$$

$$\angle SBC = 180^\circ - \theta$$

$$\angle ABS = \theta$$

$$\text{In } \triangle BSC, (SC)^2 = a^2 + h^2 \cot^2 \beta - 2ah \cot \beta \cos(180^\circ - \theta)$$

$$h^2 \cot^2 \gamma = a^2 + h^2 \cot^2 \beta + 2ah \cot \beta \cos \theta \quad \text{--- (1) } \checkmark$$

$$\text{In } \triangle ABS, h^2 \cot^2 \alpha = a^2 + h^2 \cot^2 \beta - 2ah \cot \beta \cos \theta \quad \text{--- (2) } \checkmark$$

$$\text{Adding, (1) + (2)}$$

$$h^2 (\cot^2 \alpha + \cot^2 \gamma) = 2a^2 + 2h^2 \cot^2 \beta \quad \checkmark$$

$$R^2 (\cot^2 \alpha + \cot^2 \gamma - 2 \cot^2 \beta) = 2a^2$$

$$h = \frac{a\sqrt{2}}{[\cot^2 \alpha + \cot^2 \gamma - 2 \cot^2 \beta]^{\frac{1}{2}}} \quad \checkmark$$

### Question 6

(a) a, b, c are roots of  $x^3 + qx + r = 0$

$$\text{let } y = b+c-a$$

$$= (b+c+a) - 3a$$

$$= 0 - 3a$$

$$\therefore a = -\frac{y}{3} \quad \checkmark$$

$$\left(-\frac{y}{3}\right)^3 + q\left(-\frac{y}{3}\right) + r = 0 \quad \checkmark$$

$$y^3 + 9qy - 27r = 0 \quad \checkmark$$

(b) (i)  $\tan 3\theta = \tan(2\theta + \theta)$

$$\text{RHS} = \frac{\tan 2\theta + \tan \theta}{1 - \tan 2\theta \tan \theta} \quad \checkmark$$

$$= \frac{\frac{2 \tan \theta}{1 - \tan^2 \theta} + \tan \theta}{1 - \frac{2 \tan \theta}{1 - \tan^2 \theta} \tan \theta}$$

$$= \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \quad \checkmark$$

(ii) let  $\theta = \tan^{-1} x$

$$\therefore \tan(3 \tan^{-1} x) = \frac{3x - x^3}{1 - 3x^2}$$

$$\text{using } 3 \tan^{-1} x = \frac{\pi}{2} - \tan^{-1}(3x) \quad \checkmark$$

$$\tan(3 \tan^{-1} x) = \tan\left(\frac{\pi}{2} - \tan^{-1}(3x)\right) \quad \checkmark$$

$$= \cot(\tan^{-1}(3x))$$

$$= \frac{1}{\tan[\tan^{-1}(3x)]} \quad \checkmark$$

$$\frac{3x - x^3}{1 - 3x^2} = \frac{1}{3x}$$

$$3x^4 - 12x^2 + 1 = 0 \quad \checkmark$$

$$x^2 = \frac{12 \pm \sqrt{144 - 12}}{6}$$

$$\therefore x = 0.292 \text{ only } \checkmark$$

$$x = 1.979 \text{ is not the solution.}$$

(c) (i)  $\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \dots$

$$+ \frac{1}{(n+1)(n+2)(n+3)}$$

$$= \frac{(n+1)(n+4)}{4(n+2)(n+3)}$$

When  $n=0$ , LHS =  $\frac{1}{6}$ , RHS =  $\frac{1}{6} \quad \checkmark$

Assume that it is true for  $n=k$

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \dots + \frac{1}{(k+1)(k+2)(k+3)}$$

$$= \frac{(k+1)(k+4)}{4(k+2)(k+3)} \quad \checkmark$$

$$\text{RTP: } S_{k+1} = S_k + T_{k+1}$$

$$= \frac{(k+2)(k+5)}{4(k+3)(k+4)}$$

$$S_{k+1} = \frac{1}{(k+2)(k+3)(k+4)} + \frac{(k+1)(k+4)}{4(k+2)(k+3)}$$

$$= \frac{4 + (k+1)(k+4)(k+4)}{4(k+3)(k+4)(k+2)}$$

$$= \frac{(k+1)(k^2 + 8k + 16) + 4}{4(k+2)(k+3)(k+4)}$$

$$= \frac{(k+2)^2(k+5)}{4(k+2)(k+3)(k+4)}$$

$$= \frac{(k+2)(k+5)}{4(k+3)(k+4)} \quad \checkmark$$

Since it is true for  $n=0$ ,

it is proven true for  $n=k+1$ .

$\therefore$  it is true for  $n=0+1=1$

true for  $n=1+1=2$

$\therefore$  it is true for all  $n \geq 0 \quad \checkmark$

$$(ii) \lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{1}{(n+1)(r+2)(r+3)}$$

$$= \frac{(n+1)(n+4)}{4(n+2)(n+3)}$$

$$= \frac{(\frac{n}{n} + \frac{1}{n})(\frac{n}{n} + \frac{4}{n})}{4(\frac{n}{n} + \frac{2}{n})(\frac{n}{n} + \frac{3}{n})} \quad \checkmark$$

$$= \frac{1 \times 1}{4 \times 1 \times 1} = \frac{1}{4} \quad \checkmark$$

### Question 7

$$\begin{aligned} \text{(a) (i)} \quad I_n &= \int_0^\pi \sin^n \theta \, d\theta \\ &= \int_0^\pi \sin \theta \cdot \sin^{n-1} \theta \, d\theta \\ &= [-\cos \theta \cdot \sin^{n-1} \theta]_0^\pi - \int_0^\pi \cos \theta \cdot (n-1) \sin^{n-2} \theta \, d\theta \\ &= 0 + (n-1) \int_0^\pi \cos \theta \sin^{n-2} \theta \, d\theta \\ &= (n-1) \int_0^\pi (1 - \sin^2 \theta) \sin^{n-2} \theta \, d\theta \\ &= (n-1) \int_0^\pi \sin^{n-2} \theta \, d\theta - (n-1) \int_0^\pi \sin^n \theta \, d\theta \\ &= (n-1) I_{n-2} - (n-1) I_n \\ n I_n &= (n-1) I_{n-2} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{(i)} \quad I_8 &= \frac{7}{8} I_6 \\ &= \frac{7}{8} \cdot \frac{5}{6} \cdot I_4 \\ &= \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot I_2 \\ &= \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot I_0 \\ &= \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \pi \\ &= \frac{35\pi}{128} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \int_0^\pi x \sin^n x \, dx &= \int_0^\pi (\pi - x) \sin^n (\pi - x) \, dx \\ &= \int_0^\pi (\pi - x) \sin^n x \, dx \\ &= \pi \int_0^\pi \sin^n x \, dx - \int_0^\pi x \sin^n x \, dx \\ 2 \int_0^\pi x \sin^n x \, dx &= \pi \int_0^\pi \sin^n x \, dx \\ \therefore \int_0^\pi x \sin^n x \, dx &= I_n \left( \frac{\pi}{2} \right) \quad \checkmark \end{aligned}$$

$$\text{(b)} \quad x = \cos \theta + i \sin \theta$$

$$\text{(i)} \quad \frac{1}{x} = \cos \theta - i \sin \theta$$

$$\therefore x + \frac{1}{x} = \cos \theta + i \sin \theta + \cos \theta - i \sin \theta = 2 \cos \theta$$

$$\begin{aligned} x^n + \frac{1}{x^n} &= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta \\ &= 2 \cos n\theta \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad 5x^4 - 11x^3 + 16x^2 - 11x + 5 &= 0 \\ 5\left(x^2 + \frac{1}{x^2}\right) - 11\left(x + \frac{1}{x}\right) + 16 &= 0 \quad \checkmark \end{aligned}$$

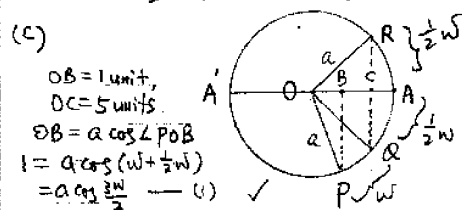
$$5 \cdot 2 \cos^2 \theta - 11 \cdot 2 \cos \theta + 16 = 0$$

$$10 \cos^2 \theta - 11 \cos \theta + 3 = 0 \quad \checkmark$$

$$(5 \cos \theta - 3)(2 \cos \theta - 1) = 0$$

$$\begin{aligned} \cos \theta &= \frac{3}{5} \text{ or } \cos \theta = \frac{1}{2} \\ x &= \frac{3}{5} + \frac{4}{5}i, \quad x = \frac{3}{5} - \frac{4}{5}i \quad \checkmark \end{aligned}$$

$$x = \frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad x = \frac{1}{2} - \frac{\sqrt{3}}{2}i \quad \checkmark$$



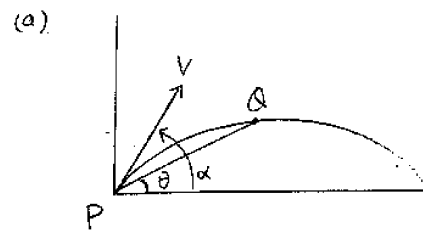
$$\begin{aligned} OC &= a \cos \angle COA \\ 5 &= a \cos \frac{\omega}{2} \quad \checkmark \end{aligned}$$

$$\text{(a)} \quad 5 = \frac{\cos \frac{1}{2} \omega}{\cos \frac{3}{2} \omega}$$

$$\begin{aligned} 5 \cos 3\left(\frac{\omega}{2}\right) &= \cos \frac{1}{2} \omega \\ 5 \left[ 4 \cos^3 \left(\frac{1}{2} \omega\right) - 3 \cos \frac{1}{2} \omega \right] &= \cos \frac{1}{2} \omega \\ \cos^2 \left(\frac{1}{2} \omega\right) &= \frac{16}{25} \Rightarrow \cos^2 \left(\frac{\omega}{2}\right) = \frac{4}{5} \\ \cos \omega &= 2 \cos^2 \frac{1}{2} \omega - 1 \\ &= -\frac{3}{5} \quad \checkmark \end{aligned}$$

$$\text{Period } T = \frac{2\pi}{\omega} = \frac{2\pi}{\cos^{-1} \left( \frac{3}{5} \right)} \quad \checkmark$$

### Question 8



Let  $x$  and  $y$  be the horizontal and vertical components of velocity when the ball is at Q.

$$\begin{aligned} \dot{x} &= V \cos \alpha, \quad \dot{y} = V \sin \alpha - gt \\ \tan \theta &= \frac{y}{x} = \frac{V \sin \alpha - \frac{1}{2} g t^2}{V \cos \alpha} \quad \checkmark \end{aligned}$$

$$\therefore \tan \theta = \tan \alpha - \frac{gt}{2V \cos \alpha}$$

The direction of motion at Q is inclined to the horizontal at  $\tan^{-1} \left( \frac{y}{x} \right) = \tan^{-1} \left[ \tan \alpha - \frac{gt}{V \cos \alpha} \right] \quad \checkmark$

$$\begin{aligned} \text{But } \frac{gt}{V \cos \alpha} &= 2 \tan \alpha - 2 \tan \theta \\ \therefore \tan^{-1} \left( \frac{y}{x} \right) &= \tan^{-1} [ \tan \alpha - 2 \tan \alpha + 2 \tan \theta ] \\ &= \tan^{-1} [ 2 \tan \theta - \tan \alpha ] \quad \checkmark \end{aligned}$$

$$\text{(b) (i)} \quad x = V \cos \theta \cdot t$$

$$y = V \sin \theta \cdot t - \frac{1}{2} g t^2$$

$$\text{Time of flight } t \Rightarrow y = 0 \Rightarrow t = \frac{2V \sin \theta}{g} \quad \checkmark$$

$$\therefore \text{Range} = V \cos \theta \times \frac{2V \sin \theta}{g}$$

$$R = \frac{V^2 \sin 2\theta}{g} \quad \checkmark$$

$$\text{(ii)} \quad R - P = \frac{V^2 \sin 2\alpha}{g} \quad \text{--- (1) } \checkmark$$

$$R + Q = \frac{V^2 \sin 2\beta}{g} \quad \text{--- (2) } \checkmark$$

$$\frac{V^2 \sin 2\theta}{g} - P = \frac{V^2 \sin 2\alpha}{g}$$

$$\frac{V^2 \sin 2\theta}{g} + Q = \frac{V^2 \sin 2\beta}{g}$$

$$\frac{P}{Q} = \frac{\sin 2\theta - \sin 2\alpha}{\sin 2\beta - \sin \theta} \quad \checkmark$$

$$(P+Q) \sin 2\theta = P \sin 2\beta + Q \sin 2\alpha$$

$$\sin 2\theta = \frac{P \sin 2\beta + Q \sin 2\alpha}{P+Q} \quad \checkmark$$

$$\text{(c) (i)} \quad \left( \frac{1}{4} \right)^3 = \frac{1}{64} \quad \checkmark$$

$$\text{(ii)} \quad \sum_2 \left( \frac{3}{4} \right)^2 \left( \frac{1}{4} \right)$$

$$= \frac{27}{64} \quad \checkmark$$

$$\text{(iii)} \quad \sum_2 \left( \frac{1}{4} \right)^2 \left( \frac{1}{4} \right) + \sum_3 \left( \frac{1}{4} \right)$$

$$= \frac{27}{32} \quad \checkmark$$

$$\text{(iv)} \quad \sum_2 \left( \frac{1}{4} \right) \left( \frac{1}{4} \right) \left( \frac{2}{4} \right) \quad \checkmark$$

$$= \frac{3}{32} \quad \checkmark$$

— END OF PAPER —